# WUYUE LI

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## EDUCATION BACKGROUND

**University of Macau** **Macau SAR**
*Bachelor of Science in Computer Science* 09/2022-06/2026

- **Overall GPA**: 3.55/4.0
- **Honors**: Dean list (2023-2026);
- **Computer Skills**: C, C++, Python, Java, SPSS
- **IELTS: 7** (R7.5/L7.5/S6.0/W6.0)
- **Activities**: Deputy Director of the Journal Studio, China Mainland Students' Association, UMSU (03/2023-04/2024); Publicity Manager, Macau University Survival Guide (Student-run Organization)(12/2022-12/2024)
- **Hobbies:** Piano, Chinese painting, Calligraphy

## PUBLICATION

**Li, W.** (2026). "Fake News Detection: A Review of Machine Learning and Deep Learning Methods." *4th International Conference on Computer, Machine Learning and Artificial Intelligence* (CMLAI), San Francisco, CA, USA. (Accepted for publication)

## INTERNSHIP ECXPERIENC

**Companhia de Electricidade de Macau (CEM)** **Macau SAR**
*Summer Trainee* 06/2025- 07/2025

- Normalized database to remove 340+ duplicate CMDB records and improved Python-to-Golang workflows.
- Followed Agile methodologies to deliver 26 project on schedule, managing 15% scope changes during the CMDB migration.
- Assisted the team with daily tasks, projects, and operations, including administrative work, research, assignments, and reports, while gaining knowledge of industry practices and processes.
- Joined a CEM site visit to gain hands-on experience with frontline power system operations.
- Explored multiple departments (CSD, PLD, GED, PND, TAD, ISD, HRS) to build understanding of businesses and operations.

**Oriental Fund** **Beijing, China**
*Intern, Department of Electronic Technology* 06/2024- 08/2024

- Observed and learned the use of office software (OA, WPS) and gained familiarity with daily IT operations.
- Shadowed the Electronics & Technology Department in resolving IT, hardware/software, and access issues, and learned to coordinate with outsourcing companies.
- Gained exposure to stock market monitoring and related operational practices, developing knowledge of financial industry.
- Enhanced practical hardware skills through hands-on learning and technical support observation.
- Participated in AIR Summer Camp 2024 on LLM system optimization; gained hands-on experience with modern LLM architectures, Python frameworks, llama.cpp, and RAG techniques.
- Proposed an activation transposition method to enable weight reuse and parallel computation for high-dimensional activations.

## RESEARCH EXPERIENCE

**Compositional Reasoning with Tricks** **Macau SAR**
*Core Member, NLP2CT Lab, Advisor: Dr. Derek F. Wong* 11/2023-present

- Built a Python-based dataset framework, including full implementation code, to support large-scale math problem datasets.
- Collected and curated datasets of math problems with step-by-step solutions and developed accompanying theoretical content.
- Explored evaluation methods for large models and techniques to measure mathematical expression difficulty.
- Planned to apply Reinforcement Learning and Direct Preference Optimization to enhance accuracy while reducing complexity and cost.

**Visaidmath: Benchmarking Visual-aided Mathematical Reasoning** **Macau SAR**
*Core Member, NLP2CT Lab, Advisor: Dr. Derek F. Wong* 12/2022-02/2023

- Searched for and collected relevant data sources to support dataset construction.
- Converted math problems from PDF files into structured JSON format.
- Integrated converted data into the existing dataset framework for further processing.
- Labeled the data within the framework to ensure clarity and usability and delivered processed and labeled datasets.

## PROJECT EXPERIENCE

**Introductory Programming Data Structures** **Remote**
*Online Training Program, supervised by Prof. Gregory Kesden, Carnegie Mellon University* 07/2025-08/2025

- Studied fundamental and advanced data structures and algorithms, focusing on correctness, performance, and optimal use.
- Presented arguments for the correctness or incorrectness of given algorithms, applying rigorous logical reasoning.
- Analyzed and evaluated algorithmic efficiency using theoretical and empirical approaches.
- Chose appropriate data structures and algorithms to solve diverse computational problems.
- Implemented high-quality algorithms and data structures with emphasis on robust, efficient code design.
- Applied algorithmic principles to program design in the broader context of machine learning and artificial intelligence.

**CISC3014 Information Retrieval and Web Search Final Project** **Macau SAR**
*Team Member, Advisor: Dr. Pengyang Wang* 11/2024-12/2024

- Processed Xiaohongshu trending titles and content using TF-IDF and Random Forest.
- Evaluated the effectiveness of data processing methods through systematic performance assessment.
- Enhanced proficiency in Python by developing data processing frameworks and handling structured data with JSON.

**CISC3019-Computer Vision Final Project** **Macau SAR**
*Team Member, Advisor: Dr. Liming Zhang* 10/2024-11/2024

- Applied Gaussian Pyramid and Laplacian Pyramid algorithms to process multiple images according to project requirements.
- Implemented image collage by extracting a region from one image and blending it into another to create visual illusion effects.
- Developed code for Laplacian Pyramid construction and region blending, ensuring smooth transitions between images.